TITLE OF THE PROJECT

WASH-HPV: The role of Water, Sanitation & Hygiene on HPV infection among women in Bondo Sub-County, Siaya County, Kenya

INVESTIGATORS AND INSTITUTIONAL AFFILIATION

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Host Institution: Jaramogi Oginga Odinga University of Science and Technology

Study Site

Bondo sub-county, Siaya County Kenya

Duration

24 months from ethical approval to final report submission

Funding Institution: VLIR-UOS BELGIUM

PROJECT SUMMARY

Persistent infection with the Human Papillomavirus (HPV) is the causative agent of nearly all cases of cervical cancer. The burden of cervical cancer in Kenya remains significant with an estimated 3250 deaths annually. HPV is a sexually transmitted infection. Maintaining good hygiene is crucial for women in prevention and reduction of HPV infection. Yet, in Kenya, according to the WHO UNICEF Joint Monitoring Programme (JMP), only 38% of households have basic hygienic conditions. Access to drinking water and sanitation are preconditions to create hygienic conditions. Despite the importance of hygiene in reducing HPV infection, the connection between HPV infection and the access to drinking water, sanitation and hygiene (WASH) is underexplored. In the WASH-HPV project, we will simultaneously research the HPV and other Sexually transmitted infections(STIs), WASH facilities and drinking water quality in a household. Specifically, 1) The study will assess the access and functional use of WASH facilities for full body and genital washing among women in Bondo 2) Analyze water quality of the available water in households of women in Bondo town 3) Determine prevalence of HPV infection among women in Bondo 4) Determine the association of WASH factors and HPV infection among women in Bondo. We will employ quantitative method approach including household survey among women(n=422). The findings will inform interventions to better integrate WASH in the prevention of HPV infection.

LAY SUMMARY

Cervical cancer is one of the life threatening diseases causing the death of many women across the world. It has been shown that infection with human papillomavirus is the cause of cervical cancer. HPV is usually transmitted by contact especially during sexual activity. It is therefore thought that the aspect of maintaining cleanliness among women could prevent and reduce the infection with HPV. Cleanliness, entails the need for women to maintain hygiene as it relates to their sexual life. However, there are limited studies that have been done to explore how aspects of hygiene which include availability of water, sanitation facilities and usage can be linked to HPV infection. For instance, do women in Bondo have access to sufficient water? do they use the water to take care of their body cleanliness before and after sex. Is there a connection that women who maintain cleanliness lower the risk of HPV infection and other STIs? Therefore, this study will explore the relationship between water, sanitation and hygiene and HPV infection. The study will involve a community household survey involving 422 women. The study has sections, firstly, each woman will have an opportunity to take part in an interview that relate to their socio-demographics, sexual health history, water, hygiene and sanitation aspects. We will do water quality testing and the women will have a chance to give their cervical swab sample for further analysis. The findings are important in helping understand the possible link that exists between water, hygiene, sanitation and HPV infection.

BACKGROUND INFORMATION AND RATIONALE

The project aims to address the burden of HPV infection and subsequent cervical cancer caused by limited access to water, sanitation and hygiene. Cervical cancer, a preventable disease, is a significant public health threat, with 604,000 new cases globally and 3,250 deaths annually among women in Kenya(1).It is considered as the major cause of cancer related deaths among women worldwide. It is essential to note that the world largest cervical cancer burden is reported in Sub-Saharan Africa with Kenya being part of 19 countries with the greatest burden. In 2020, WHO launched the global cervical cancer elimination strategy identifying key interventions whose implementation would lead to the elimination of cervical cancer. The strategy also referred to as the 90:70:90 cascade projects that by 2030, 90% of girls should be fully vaccinated against HPV by the age of 13; 70% of women screened with a high precision test at least twice before age 30 and 40 years; 90% of women with pre-cancer and invasive cervical cancer should receive comprehensive care and treatment. Kenya is currently not meeting the WHO targets. Furthermore, the cascade does not consider the connection and probable link that WASH factors could have in contributing to HPV infection.

Persistent infection with HPV has been documented as the causative agent of cervical cancer. HPV is an ubiquitous virus that is mainly transmitted by close skin contact and especially through sexual contact. The process begins when HPV infects basal epithelial cells of the cervix, typically occurring at the transformation zone, often coming after micro-trauma that exposes these cells. It is essential to note that most HPV infections are transient and often clear on their own by the immune system. It is the persistent infection with high risk HPV that is necessary for carcinogenesis. HPV infection remains common in western Kenya. A 2023 study in Bondo (The pilot region of this project) reported a 41.6% HPV prevalence among women, exceeding the national average(2). Similarly, another study that was to determine HPV prevalence among HIV positive and HIV Negative women in Eastern Kenya revealed a prevalence of 27.6%. Evidently, HPV infection remains common in Kenya.

Nevertheless, the linkage between WASH factors and HPV infection has not been extensively studied in medical literature. Recent studies provide indirect evidence relevant to this question. In a previous study that was done to monitor the presence of HPV in public areas indicated the possibility of non-sexual transmission of the virus and environmental risks. HPV was detected in public restrooms. Health centres and even hospitals with highest detection rates on squat toilets and washing basins(3). There has been inconclusive findings on the role of washing of genitalia and HPV infection among women. For instance, there are studies that have indicated that douching could be associated with adverse health outcomes including sexually transmitted bacterial infection and cervical cancer(4). On the other hand, some studies suggest that douching reduces the likelihood of genital warts, type 6 and 11 of HPV infection(5). Hypothetically, douching especially after sexual intercourse have been considered to help clear transmitted HPV. A study was done among 200 female sex workers in Cambodia to explore intravaginal practices and HPV infection(6). The findings revealed that a lower number of infecting HPV genotypes were associated with intravaginal washing in the past 3 months(6). These findings challenged some of the existing views that all types of vaginal cleansing were harmful. Specifically, intravaginal washing specifically using water shortly after sex would help prevent HPV infection among female sex workers.

Despite the indirect evidences presented, in Kenya, the problem of water scarcity has been on the rise. In fact, it is considered one of the water scares countries across the world with per capita availability of below 1000m annually. Research shows that the struggle for access of clean and safe water is a problem experienced by more than 18 million people. Previous studies further reveal that only about 56% of the population has access to clean water(7). Citizens mainly in rural and slums settings of urban areas travel up to long distances to reach water zones or have to wait for hours and days to get water due to water rationing(7).

As mentioned, maintaining good hygiene is crucial for prevention and reduction of HPV, but in Kenya, only 38% of households have basic hygiene (handwashing facility with water and soap) per the 2022 WHO UNICEF JMP.(8) 28% have water but no soap, and 34% lack a handwashing facility altogether. While drinking water access is better (63%), sanitation access remains low (37%)(8). Despite the importance of genital washing for preventing HPV transmission, it's not specifically monitored. While 89% of Kenyans report having a private place to wash at home, this contrasts sharply with the low overall access to basic hygiene, suggesting a potential gap in effective HPV prevention.

Further, globally, but also in Kenya, data, policies and actions on reducing HPV infection are separated from those of WASH despite evident connections. The existing policies, at global and Kenyan level, illustrate that both domains, i.e. cervical cancer and WASH are developed as standalone silos. Kenya aims to eliminate cervical cancer by 2030, mirroring the WHO's global strategy, and also strives for universal access to sanitation and hygiene through the Kenya Environmental Sanitation and hygiene policy(1, 9). While both national strategies (Cancer Control Strategy and KESHP) are led by the Ministry of Health, they surprisingly fail to connect hygiene with HPV infection risk, despite growing evidence suggesting a link between poor WASH conditions and HPV. This disconnect highlights the need for further research and integrated strategies.

This project, addressing the WASH-HPV link, directly supports SDG 3 (health and wellbeing), particularly target 3.4 (reducing non-communicable diseases like cervical cancer). It also champions gender equality (SDG 5)sand access to clean water and sanitation (SDG 6), while prioritizing equity and women's rights in Kenya(1)

In the project, we will simultaneously assess the access and functional use of WASH facilities, especially for full body and genital washing among women, while also analysing the HPV and other STIs infection rate through a cervical swab, blood samples and analysing the water quality of the available water in Bondo town, western Kenya (by means of chemical and bacteriological analysis) through partnership with community Health promoters, Local county government-Ministry of Health, Ministry of Water, University of Antwerp and JOOUST. .

PROJECT OBJECTIVES

Primary Objectives

To assess the the role of Water, Sanitation & Hygiene on HPV infection among women in Kenya

Secondary Objective

1. Assess the access and functional use of WASH facilities for full body and genital washing among women in Bondo sub county
2. Analyze water quality of the available water in households of women in Bondo sub-county
3. Determine prevalence of HPV infection among women in Bondo sub-county
4. Determine the prevalence of other STIs (Trichomonas Vaginalis, Chlamydia Trachomatis, syphilis, Neisseria gonorrhoeae, Hepatitis B) among women in Bondo sub county
5. Determine the association of WASH factors and HPV infection among women in Bondo sub-county

METHODOLOGY

The scope

The WASH-HPV project is a cross-sectional study that will be undertaken within two years to explore the role of water, hygiene and sanitation on HPV infection among women in Kenya. Specifically, the WASH-HPV project will be undertaken in Bondo town, Siaya County, Western Kenya. Bondo has five wards that make up the municipality. The sample will be drawn from each ward based on stratified sampling. The population of the town is linear with few elements of scattered population. The target population include adult women and living within Bondo town. Each woman must meet the inclusion criteria which includes being of reproductive age, no history of cervical cancer or hysterectomy and belongs to a household. They will be recruited after the mapping of the households. The key activities include community outreach to create awareness on HPV infection and cervical cancer burden. Other activities include household survey, HPV sampling, assessment of WASH factors, Water quality testing, HPV and other STIs analysis, bacteriological analysis of water, data entry and analysis, report, manuscript writing and publication and presentation of findings. The main variables under investigation in this project include HPV infection and WASH factors. Other variable includes STI infections. The variables will be investigated separately then combined in a regression analysis to make inferential findings. The main limitation of the study is that it is cross- sectional and therefore it is not possible to follow up women over a period of time and establish how changes in WASH factors influence HPV infection rate.

Study Site

The study will be conducted in Bondo sub-county located in Siaya County which is approximately 64KM west of Kisumu, the second largest city of Kenya. From the last census, the region has an estimated population of 22712. Recent report in 2023 reveal that 57% of people living with HIV in Kenya come from western Kenya of which is Siaya county that houses Bondo municipality is among them. HIV is a major risk factor for HPV infection. Women who are HIV positive are 6 times more likely to have HPV infection compared to HIV negative women. This justifies the choice for the study site.

Study design

This is a cross-sectional study to be undertaken in the community. The project shall begin by mapping target households within Bondo sub county. Subsequently, the research assistant will visit the identified households on a day to day basis until the sample size is achieved. During the visit, the researchers will administer questionnaires and carry out interviews with 422 women belonging to each household within Bondo municipality. This sample size was arrived at using Fisher's formula for population of over 10000. To ensure equal distribution within the five wards, proportionate stratified random sampling will be employed. This means that Bondo Municipality will be divided into its 5 wards and sample population stratified according to the population of each ward. Finally, random sampling will be utilized in each stratum because the exact number of households is not known. Each household to be included in the study will have to meet an inclusion criterion of having a woman of reproductive age as a member of the household. The participants shall only be take part after undergoing a recruitment process that involves explaining the project in details, written informed consent. We will collect swabs, urine

Inclusion criteria

To be eligible for participation in the study, women must meet all of the following criteria

- They must be adult women from late adolescent
- They must be residing in the selected study location
- Willingness to provide consent

Exclusion criteria

- Pregnant women will not take part in the study
- History of total hysterectomy
- Prior diagnosis or treatment of cervical precancerous/cancer
- Mental impairment that in opinion of research team is unable to provide study procedures

Data Collection Procedures

Interviewers Training

The project plans to employ female research assistant who are fluent in Luo dialect as this will be critical in administration of the questionnaires and interviews. Bondo is largely made of Luo speaking community with strong cultural orientation where women embrace discussing sexual reproductive health issues with fellow women. The research assistants who will be the interviewers in this project will undergo a five-day training. The training will focus on 5 important areas including: project immersion and ethical foundation, mastering the questionnaire content; Kobocollect proficiency; Interviewing skills and practice; Field logistics and pilot testing.

Pilot Study

A pilot study will be conducted with approximately 42-50 women (10% of the study population) in a non-study area within Bondo sub county to pre-test the interview tool, consent process, sample collection procedures and logistical flow. Feedback will be used to refine the protocol and tools as well.

Community Mobilization

In collaboration with community health promoters and local leaders, awareness campaigns will be conducted in the selected wards to explain the study purpose and benefits thus promoting community acceptance and participation. The project will begin by carrying out a community entry meeting and County stakeholder meeting. The County stakeholder meeting will come first and aims at bringing the local county leaders and healthcare providers on the objective of the project and creating awareness of HPV infection and potential linkage to WASH. The County stakeholder meeting will also act as a catalyst that will encourage all women to attend the subsequent community entry meeting. The community entry meeting will follow and will majorly involve community health promoters and women from Bondo Municipality. The major aim of the community entry meeting would be to provide awareness on the current state of cervical cancer in the country, the probable linkage between WASH and HPV infection and the need for further studies. In essence, this is a meeting that aims to promote awareness on HPV infection and associated risk factors and the anticipated context of WASH. The researchers will also use the meeting to explain the new project and how it will be undertaken in the community.

Main Study

We shall then proceed to collect data in two phases HPV related data and Water/Sanitation assessment Data

The research assistant by the help of community Health promoters will visit the mapped households. Each adult woman in each household will be interviewed on one on one basis. The interview will focus on demographics, reproductive health and sexual behavior, Personal hygiene practices.

Water and Sanitation Assessment Data

From each household, the researchers will collect data on water availability, access and sanitation though direct observation and interviews using a Kobo-based toolkit installed on a smart phone. Water quality data will be collected in each household using on-site water testing kits and the results reported real time in the Kobo-based tool kit. Data will be transferred by smart phones to a central project server. Additional water samples will be drawn and transported to JOOUST Lab for measurement of specific physical, chemical and biological water quality parameters (Clarity, Nitrate, Ph, temperature, electrical conductivity, dissolved oxygen, Nitrite, phosphate concentration, macro- invertebrates and E. coli).

Molecular Analysis

The RIATOL qPCR HPV assay will be used to extract and genotype HPV-DNA at JOOUST Miyandhe campus. JOOUST has newly established molecular Laboratory that is funded by the Sonic Foundation. This will be the first project to be undertaken in the Laboratory and will therefore act as a validation and lab operationalization project. The choice to use RIATOL HPV genotyping qPCR assay was decided on because of its proven clinical accuracy, reproducibility and quantifiable results as it provides viral load information. Briefly, cervical samples collected with the multi-Collect Specimen Collection Kit (Abbott) and 300 µl of each sample will be transferred into the 96-deep well. DNA will be the viral DNA/RNA extraction kit. The DNA extract will then store at 4°C awaiting PCR. As an internal cellular control, beta-globin will be employed to assess the validity of the PCR run. To achieve relative quantification of data and confirm the qPCR acceptable efficiency ranges of 90–110%, standard curves will be generated by amplification of a series of synthetic DNA ten-fold dilutions. Subsequently, 384-PCR well plates will be prepared) by mixing 2 µL of extracted DNA or specific control with 4 µL of the assigned master mix solutions, as described. In total, each sample was tested with 8 different master mixes, including primers and probes for detection of HPV 6, 11, 16, 18, 31, 33,35, 39, 45,51, 52, 53 56, 58, 59, 66, 67, 68 and beta-globin. RT-amplification will be carried out on the Light Cycler 480 (Roche) instrument.

Nucleic acid amplification tests will be carried to determine the specific STIs among the women according standard operating procedures.

Bacteriological Water Analysis

Water samples will be incubated in specific growth media to encourage bacterial growth. Usually, different media are used to target specific bacteria types. The research team members will be keen to identify the presence of coliform bacteria, E. Coli, salmonella among others.

Data collection will be undertaken using Kobo collect tool for quantitative data. This will involve creation of survey forms and training of research assistants on the use of the Kobo collect app, questionnaire administration, informed consent and ethical considerations. The project coordinator will ensure regular supervision and quality checks of data collection. There will be regular upload of the collected data from kobo collect to the kobo toolbox server. This will be followed by real time monitoring of data completeness, consistency and identification of errors in the kobo collect platform. The researchers will ensure data storage and restricted access and regular backups. All the results of the household Survey-Assessing WASH factors, women reproductive health, molecular HPV genotyping, Water quality testing and bacteriological water analysis will then be double entered into an excel sheet.

Data Analysis

The initial steps will involve comprehensive data cleaning including identification of missing values, outliers and inconsistencies. Descriptive statistics will be used to describe the sociodemographic characteristics of the study population, describe WASH indicators, and report prevalence of HPV infection and other STI infections. We will further carry out inferential statistics such as Chi-square or fisher exact test to assess association between categorical

WASH variables and HPV infection status, Logistic regression to identify independent predicators of HPV infection controlling for confounders that will help determine the contribution of WASH factors. Where appropriate, we will carry out generalized linear Models(GLM) for more complex relationships. We also plan to do spatial analysis and thus map out distribution of HPV infection and WASH indicators and also analyze spatial correlations between WASH access and HPV prevalence. Prevalence of HPV will be calculated as a proportion of study participants expressed as frequency and percentage. To check for association between WASH factors and HPV infection, regression analysis will be carried out.

Ethics

Prior to undertaking the project, we will seek for ethical approval from and JOOUST research ethics committee. The research will be licensed by National Commission for Science, Technology and Innovation and JOOUST research ethics committee and JOOTRH ethic review committee. All participants will be provided with written informed consent. The ethical standards for the regulation of research in humans in accordance with Declaration of Helsinki will be followed. Further approval from the county government of Siaya under the ministry of health will be acquired.

Dissemination of Results

The findings of the study will be disseminated through peer reviewed publications. It is expected that the research team will have at least two publications ate the end of the research period. The findings will also be presented at National HPV and WASH conferences in Kenya. We will also hold stakeholder meetings and present our report bringing along ministry of Health and Ministry of water health departments. We also look forward to having community feedback sessions with study participants and local leaders as well. We will also hold policy briefs to the relevant stakeholders based on our findings.

BUDGET

The project budget is a total Ksh. 8189,400 in order to assess the role of water, sanitation, hygiene (WASH) on HPV and other STIs infection in Kenya.

The total investment cost is Ksh. 560,000 and this majorly covers the cost of one laptop and tablets that will used for data management. Specifically, the laptop will be used for information management. The tablets will be used by the research assistant for data collection in Bondo. We have also set aside money for hiring of personnel at 350, 000. The project requires a Data tool expert, data analysis, water quality person and a laboratory technician. We plan to hold community workshops and outreaches at 1,547,000. The workshops aim at sensitization and discussion prior to and after the study. The data collection process will cost 882,000.

Operational costs will cover a significant portion of the budget totalling Ksh. 1,890,000. This involves purchase of Lab consumables, screening consumables, water quality and assessment kits, DNA extractions kits, HPV molecular kit, stationeries, and subsistence, stakeholder meetings and outreaches and publications. The dissemination process will cost 700,000 while the local and international travel cost will cost 1,300,000. Overhead costs have been set at Ksh. 904,400

A detailed table has been attached in the appendix illustrating the budget and expense lines.

Timeline

The project is expected to take maximum of 2 years beginning September 2025 to September 2027 with the first six month set for preparatory work including seeking ethical approvals, community engagements, hiring of experts, recruitment and training of research team and enrollment of women. The next 6 months will involve piloting the study and actual study. The first 6 months of the second year will involve laboratory and molecular analysis of the samples. The final six months will involve report writing, development of manuscript and dissemination of the findings to the relevant stakeholders.

Table 1: Study timelines

Activities/Project Year	**Year 1: Sept 2025-Sept 2026**				**Year 2: Sept 2026-Sept 2027**			
	1	2	3	4	1	2	3	4
Institutional agreements/contracts								
Protocol development, IRB ethical approvals								
Recruitments and Training								
Piloting and reconnaissance activities								
House Hold Survey								
Household Survey								
Household survey								
Laboratory Analysis								
Laboratory Analysis								
Data cleaning and analysis								
Feedback of results								
WP1-3: Reports and manuscript writing								
WP1-3: Dissemination of study findings								

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Informed Consent

Principal Investigator: Ivy Akinyi

CO-Principal Investigators: Prof Jan Cools & Prof JP Bogers

Hello, my name is…………………….I am a research assistant working with Jaramogi Oginga Odinga University of Science and Technology(JOOUST). We are conducting a study about health in the community. The main goal of the study is to find out how water, hygiene and sanitation is linked to human papillomavirus (HPV) infection and ultimately cervical cancer. Your participation in this study is completely voluntary. Before you decide to participate, I will explain this study to you. Please feel free to ask any questions you have as we go along

Purpose

The purpose of this study is to understand more about water, sanitation and hygiene practices in your household and community and how they might be linked to human papillomavirus (HPV). HPV is a common virus-transmitted by contact-that often clears on its own but persistent infection can lead to certain cancers such as cervical cancer. We hope that understanding these connection, we can develop better public health programs to improve health in the community

What will happen if I agree to participate?

If you agree to participate, we will ask you a series of a questions in a private setting. This interview will take approximately 30-45 mins and will cover the following topics

a. Socio-demographics and sexual reproductive health
b. Your household and personal use of water
c. Sanitation facilities
d. Hygiene practices
e. Link between WASH and HPV

We may ask you if you are willing to provide self-collected vaginal swab/Nurse assisted swab collection. For nurse assisted swab collection, the sample will be collected by a trained registered nurse and will be tested for HPV infection. We will explain the process of sample collection separately and ensure it is done comfortably and privately. We will also inform you about the results of HPV tests in a confidential manner

Further, we will request to sample the water in your home and urine and test for physical, biological and chemical components. We will also pick another water sample for further laboratory analysis.

Confidentiality

We will do our best to keep all the information that you have provided confidential.

- Your name will not be recorded in the questionnaire. Instead, we will use a unique code to identify your information
- All information collected will be stored securely on password-protected devices and servers accessible to authorized team only
- Any reports or publications from this study will present findings in a way that no individual can be identified
- Your biological sample will also be labelled with a unique code. The sample will be processed and stored securely at Jaramogi Oginga Odinga University of Science and Technology (JOOUST) molecular and diagnostic Laboratory in Miyandhe campus for the purpose of research only.

Benefits of Participation

Directly, there may be no immediate personal benefits to you for taking part in this study. However, the information we collect will help us

- Understand the factors that influence HPV infection
- Inform ministry of health ministry of water and other organizations on how to develop more effective programs for improving WASH practices and preventing HPV infection
- You will receive HPV results which will help you make informed decisions about your health and seek appropriate follow up if need be

Risk of Discomforts

There are minimal risks associated with participating in this study;

- The interview will take some of your time
- Sensitive questions: Some questions, especially those related to your sexual health or personal hygiene might be private and cause discomfort. You are not obligated to answer any question if you do not wish to answer.ou can stop the interview at any given time
- there might be minor discomfort during sample collection.

Voluntary participation and right to Withdraw

Your participation in this study is completely voluntary. You have the right to refuse to participate or withdraw from the study at any time for any reason without penalty or loss of benefits to which you are otherwise entitled. Your decision will not affect your access to services at Jaramogi Oginga Odinga University of Science and Technology (JOOUST) or any other facility. If you choose to withdraw, any information collected from you up to that point will be kept confidential and will not be included in the study analysis.

Contact information

If you have any questions about this study, you can contact the principal investigator: IVY AKINYI (+254729201685)

If you have any concerns about your rights as a research participant, you can contact Jaramogi Oginga Odinga University of Science and Technology (JOOUST) ethical review committee

Ethical committee contact

Consent

I have read (or had read to me) the information in this consent form. I have had opportunity to ask questions and my questions have been answered to my satisfaction. I understand that my participation is purely voluntary and that I can withdraw at any time. I understand that the information I provide will be kept confidential

I agree to participate in this research study

Participant's Confirmation of Consent:

(Interviewer to read this statement aloud and confirm understanding)

"Do you understand the study as it has been explained to you? Do you agree to participate in this study?"

- **If Yes:**
  - Participant's Name (Printed): ____
  - Participants contact
  - Participant's Signature: ______________________________
  - Add thumb option
  - LAR ____
  - Date: ___________________________
- **If No (or if participant prefers not to sign/thumbprint, record refusal):**
  - Interviewer records: "Participant declined to participate" or "Participant provided verbal consent but declined signature/thumbprint."

Interviewer's Confirmation:

I have accurately read and explained the contents of this consent form to the participant. I have ensured that the participant understands the study and has had the opportunity to ask questions, which I have answered to the best of my ability. I confirm that the participant voluntarily agreed to participate.

- Interviewer's Name (Printed): ____
- Interviewer's Signature: ____
- Date: ___________________________

WASH HPV

Questionnaire: WASH-HPV Study

Study Title: The role of Water, Sanitation & Hygiene on HPV infection among women in Bondo Sub-County, Siaya County, Kenya

Introduction: Thank you for agreeing to participate in this study. The purpose of this questionnaire is to collect information about your socio-demographics, household water, sanitation, and hygiene practices, as well as the history of your reproductive health, sexual behavior, and possible linkage to HPV infection. Your responses are very important for understanding how these factors might be related to HPV infection. All information you provide will be kept strictly confidential and used only for research purposes. There are no right or wrong answers, please answer as honestly as possible. Your participation is voluntary, and you can choose to withdraw at any time.

Instructions:

- Please answer all questions to the best of your knowledge.
- If you are unsure about a question, please ask the research assistant for clarification.

Part 1: Socio-Demographic Information

1. **Date of Interview:** ___/___/_____(DD/MM/YYYY)
2. **Interviewer ID:** ________
3. **Respondent ID:** ________
4. **Ward:** ____________________
5. **Village/Community:** _____________________
6. **Age (in years):** ________
7. **Marital Status:**
    - o Single
    - o Married/Living with a partner
    - o Divorced/Separated
    - o Widowed
8. **Highest Level of Education Completed:**

- No formal education
- Primary school (some years)
- Primary school (completed)
- Secondary school (some years)
- Secondary school (completed)
- Tertiary/Vocational training
- University/Higher education

9. **Primary Occupation:**
    - Farming/Agriculture
    - Fishing
    - Small business owner/Trader
    - Formal employment (e.g., teacher, civil servant)
    - Casual labor
    - Housewife/Homemaker
    - Student
    - Unemployed
    - Other (specify): ____________________________
10. **How many children do you have?**
    - ………………………………….

Part 2: Water Sources and Practices

1. **What is your primary source of drinking water?**
    - Borehole within your premise
    - Borehole outside your premise
    - Public county water supply (piped water)
    - Rainwater collection
    - Surface water (river, dam, lake, pond, stream, canal, irrigation channel)
    - Water vendor/Tanker
    - Other (please specify): ____________________________
2. **What is your primary source of water for other household uses like bathing?**
    - Same as drinking water source
    - Piped water
    - Rainwater collection
    - Surface water (river, dam, lake, pond, stream, canal, irrigation channel)
    - Other (specify): ___________________________
3. **How reliable is your primary source of drinking water?**
    - It is always available
    - Sometimes unavailable (a few times per week)
    - Daily water shortage issues
    - Unavailable during water scarcity periods only
4. **In case your primary water source is not available, what is your alternative**
    - I use a secondary water source (Please specify) :______________
    - I store spare water inside my premise
    - I ask the neighbors
    - Other (Please specify):
5. **How long does it typically take to fetch water (round trip, including queuing)?**
    - On premises
    - Less than 30 minutes
    - 30 minutes - 1 hour
    - More than 1 hour
6. **How safe do you think your primary drinking water source is?**
    - I believe it is safe, I never get ill from it

- I am not sure how safe it is, I assume it is fine
- I know my water is not safe, but I am strong, it doesn't affect me
- I know my water is not safe, but I cannot invest in better water
- I know my water is not safe, but I don't want to invest in better water

7. **Do you treat your drinking water at home?**
   - Yes
   - No (If No, skip to Q8)
8. **If Yes, what method do you primarily use to treat your drinking water?** (Select all that apply)
   - Boiling
   - Chlorination (e.g., WaterGuard)
   - Filtering (e.g., ceramic filter, cloth filter)
   - Solar disinfection (SODIS)
   - Settling/Decanting
   - Other (specify): ____________________________
9. If No, why don't you treat your drinking water (Select all that apply)

- I believe the water is clean
- I don't have time
- I am not aware I need to treat
- Other (specify): ____________________________
-

10. **How do you store your drinking water at home?**
    - Covered container with a narrow opening (e.g., jerrican, plastic bottle)
    - Open container (e.g., bucket, basin)
    - Water dispenser
    - Other (specify): ____________________________
11. **How often do you clean your water storage container?**
    - Daily
    - 2-3 times a week
    - Once a week
    - Less than once a week
    - Never

Part 3: Sanitation Facilities and Practices

1. **What type of toilet facility does your household primarily use?**
   - Flush/Pour-flush toilet (to piped sewer system, septic tank, or pit latrine)
   - Pit latrine with slab/covered pit
   - Pit latrine without slab/open pit
   - Composting toilet
   - No facility/Bush/Field (open defecation)
   - Other (specify): ____________________________
2. **Who is using your toilet facility??**
   - Only my household
   - It is shared with other households
   - I have no toilet facility
3. **If shared, how many households share this facility? ________**
4. Do you get in touch with faeces? (that means by bare skin touches excreta)
   - When I clean myself
   - I come in touch with the faeces of my children
   - I come in touch when I clean the latrine

- I come in touch with faeces when I want to dispose it
- I never or rarely get in direct contact with faeces

5. **Is the toilet facility clean and well-maintained?**
   - Always
   - Usually
   - Sometimes
   - Rarely
   - Never
6. **Where do you dispose of children's feces?**
   - In the toilet/latrine
   - Bury it
   - Throw in the bush/field
   - Throw in the river or lake
   - Dispose with household waste
   - Other (specify): _____
   - What menstrual hygience products do you use?
   - Sanitary towels
   - Old clothes/blankets
   - others________________________
7. **How do you manage menstrual hygiene products (e.g., sanitary pads)?**
8.
   - Dispose in the toilet/latrine
   - Wrap and dispose with household waste
   - Burn
   - Bury
   - Throw in the bush/field
   - Throw in the river or lake
   - Reusable products (e.g., cloth pads, menstrual cup) – if so, how are they cleaned and stored? ____________________________
   - Other (specify): ____________________________
9. **Do you have a designated place for bathing within your household?**
   - Yes, private indoor area
   - Yes, private outdoor area
   - Yes, shared communal bathing area
   - No, I bathe in the open/river/lake

Part 4: Hygiene Practices

1. **Is there a place for handwashing with soap and water near the toilet facility?**
   - Yes
   - No
2. **What do you typically use to wash your hands?** (Select all that apply)
   - Soap and water
   - Ash and water
   - Just water
   - Sanitizer
   - No handwashing

3. **How often is soap available?**
    - o Always, at least most of the times
    - o About 50% chance there is soap
    - o Soap is rarely available e.g. only on special occasions
    - o Soap is never available
4.
5. **At what critical times do you wash your hands with soap and water?** (Select all that apply)
    - o Before preparing food
    - o Before eating food
    - o After using the toilet/latrine
    - o After cleaning a child's bottom
    - o Before feeding a child
    - o After handling garbage
    - o After coughing/sneezing
    - o After touching animals
    - o Before and after changing menstrual hygiene products
    - o I never or rarely wash my hands with soap
    - o Other (specify): ____________________________
6. **How often do you bathe/shower?**
    - o Daily
    - o Every other day
    - o 2-3 times a week
    - o Less than 2 times a week
7. How often do you wash your hands
8. How often do you wash your genital parts
9. Daily
   after sexual intercourse
   After visiting the toilet

10. **How often do you change your underwear?**
    - o Daily
    - o Every other day
    - o 2-3 times a week
    - o Less than 2 times a week
11. **Where do you typically dispose of household solid waste (garbage)?**
    - o Collected by municipal services
    - o Buried in the yard
    - o Burned in the yard
    - o Dumped in an open space/river/lake
    - o Composted
    - o Other (specify): ____________________________
    - o ____________________________

Part 5: Health-Related Information (Self-Reported)

1. **Have you ever heard about Human Papillomavirus (HPV) before this survey?**
    - o Yes
    - o No
2. **Have you ever been screened for cervical cancer?**

- Yes
- No (If No, skip to Q35)

3. **If Yes, when was your last cervical cancer screening?**
    - Less than 1 year ago
    - 1-3 years ago
    - More than 3 years ago
    - Don't remember
4. **What was the result of your last cervical cancer screening (if known)?**
    - Normal/Negative
    - Abnormal/Positive (precancerous or cancerous changes)
    - Unclear/Inconclusive
    - Don't know/Not told
5. **Have you ever been diagnosed with HPV infection?**
    - Yes
    - No
    - Don't know
6. **Have you ever received the HPV vaccine?**
    - Yes
    - No
    - Don't know
7. **Do you have an adolescent girl age 9-14 years in your household who is your daughter?**
8. **What is the age of the girl? Please specify…………………………**
9. **Has she received HPV vaccination?**
    - Yes
    - No
10. **If yes, do you know the year she received? Specify the year**…………………………….
11. **If no, why hasn't she received HPV vaccination?**
    - I do not know about HPV vaccine
    - Concerns about safety/side effects
    - Cost of the vaccine
    - My daughter is too young
    - I don't think she needs it
    - Religious/cultural belief
    - Vaccine not available
    - Other (Please specify)………………………….

12. **Do you experience any of the following symptoms (not necessarily related to HPV, but general reproductive health)?** (Select all that apply)
    - Unusual vaginal discharge
    - Vaginal itching or burning
    - Abnormal vaginal bleeding (e.g., after sex, between periods)
    - Pain during sexual intercourse
    - Lower abdominal pain
    - Genital warts (if applicable)
    - None of the above
    - Prefer not to say

Part 5: WASH and Sexual-Reproductive Health Hygiene Practices (Sensitive Section)

Interviewer Note: Please ensure a private and comfortable setting for this section. Reiterate confidentiality and the voluntary nature of participation.

1. **Are you currently sexually active?**
    - o Yes
    - o No (If No, skip to Part 6)
    - o Prefer not to say (If Prefer not to say, skip to Part 6)
2. **How often do you typically bathe/wash your genital area?**
    - o Daily
    - o More than once a day
    - o Every other day
    - o 2-3 times a week
    - o Less than 2 times a week
3. **Do you typically bathe or wash your genital area before sexual intercourse?**
    - o Always
    - o Most of the time
    - o Sometimes
    - o Rarely
    - o Never
4. **Do you typically bathe or wash your genital area before sexual intercourse?**
    - o Always
    - o Most of the time
    - o Sometimes
    - o Rarely
    - o Never

5. **What about your sex partner, does he bathe his genital area after sexual intercourse?**
    - o Always
    - o Most of the time
    - o Sometimes
    - o Rarely
    - o Never
6. **Is your partner Circumcised?**
    - o Yes
    - o No
7. **What do you typically use to wash your genital area?** (Select all that apply)
    - o Soap and water
    - o Just water
    - o Traditional herbs/concoctions
    - o Douche products
    - o Other (specify): ____________________________
8. **If you use water for washing your genital area, what is the source of this water?**
    - o Piped water into dwelling/yard/plot
    - o Public tap/Standpipe
    - o Tube well/Borehole
    - o Protected dug well
    - o Unprotected dug well
    - o Protected spring
    - o Unprotected spring
    - o Rainwater collection
    - o Surface water (river, dam, lake, pond, stream, canal, irrigation channel)
    - o Bottled water
    - o Water vendor/Tanker

- Other (specify): ___________________________

9. **Do you or your partner(s) use condoms consistently during sexual intercourse?**
    - Always
    - Most of the time
    - Sometimes
    - Rarely
    - Never
    - Not applicable (if not in a relationship that involves condom use)
10. **How many sexual partners have you had in your lifetime?**
    - 1
    - 2-5
    - 6-10
    - More than 10
    - Prefer not to say

BUDGET

Item	**Year 1(in Ksh)**	**Year 2(in Ksh)**	**Total (in Ksh.)**
Equipment (Laptop and Tablets	560,000		560,000
Personnel	322,000	28000	350,000
Community workshop and outreaches	1,120,000	427,000	1547000
Data Collection	630,000	252,000	882000
Laboratory Analysis	1,750,000	140,000	1890000
Ethical Approvals	56000		56000
Dissemination		700,000	700,000
Overheads	490000	414,400	904.400
Travels	1,300,000		
Totals	**6,228,000**	**1,961,400**	**8,189,400**

IVY AKINYI

P.O BOX 931, KISUMU | 0729201685 | ivyakinyi44@gmail.com

PROFESSIONAL SUMMARY

I am a highly Motivated and results-oriented **early career researcher** with a robust foundation in **non-communicable disease (NCD)** research. My expertise spans the entire research lifecycle, from study design to publication. I have a proven ability to perform **complex data analysis**, interpret findings, and translate them into high-quality **manuscripts for publication**. Additionally, I am proficient in **grant writing**, successfully securing funding for research projects. My goal is to leverage these skills to contribute to innovative academic research and advance our understanding of NCDs.

KEY SKILLS

- Drafting and refining of research protocols
- Thorough knowledge of quantitative and qualitative research, advanced statistics and research methods for monitoring, including sampling techniques and the use of software for analysis (R and Stata.
- Excellent critical thinker
- Able to make strategic recommendations as part of the evaluation process.
- Excellent communication and team building skills, can establish and maintain strong working relationships inside and outside an organization.
- Manuscript Writing
- Grant writing and proposal development

EDUCATION

PHD-MEDICAL SCIENCES| SEPTEMBER 2025| UNIVERSITY OF ANTWERP-BELGIUM

Thesis: HPV IN CERVICAL DISEASE: PREVALENCE AND ROLE OF HPV-16 GENETIC VARIANT

MSC. EPIDEMIOLOGY ND BIOSTATISTICS | DECEMBER 2019 | JARAMOGI OGINGA ODINGA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thesis: Risk factors and management of outcome of Premarital sex among adolescent youths attending secondary school in Kadibo Division, Nyando Sub-County, Kisumu County, Kenya

BSC.MEDICAL LABORATORY SCIENCE | DECEMBER 2012 | KENYATTA UNIVERSITY

ADDITIONAL TRAINING

EPIDEMIOLOGY, BIOSTATISTICS AND QUALITATIVE RESEARCH (EBQ-COURSE) | OCT-NOV 2023 | ANTWERP, BELGIUM

- Basic principles and methods of epidemiology
- Medical statistics with appropriate statistical methods
- Qualitative research methods
- Systematic reviews and meta-analyses

LANGUAGES

- English- Advanced level
- Kiswahili- Advanced level
- French-Basic

EMPLOYMENT

JUNIOR RESEARCH FELLOW | JARAMOGI OGINGA ODINGA UNIVERSITY OF SCIENCE AND TECHNOLOGY- DEC 2024 TO-DATE

I. Seeking funding opportunities for projects in the School of health Science at Jaramogi Oginga Odinga University of Science and Technology (JOOUST)
II. Spearheading research projects in the school in the school of Health Science
III. Mentor and Supervise students in the school of health science
IV. Participate in School of health sciences departmental activities

RESEARCH ASSISTANT| VLIR-UOS -STRENGTHENING CAPACITY FOR DIAGNOSIS AND MANAGEMENT OF NON-COMMUNICABLE DISEASES IN LAKE VICTORIA BASIN DEC2022-DEC2023

- Project Lead Investigator in Objective 1: Molecular Characterization of Human Papilloma virus

  I. Conceptualization of study objectives
  II. Designing study protocols and tools of data collection
  III. Carrying out data analysis using R software
  IV. Manuscript writing and publication
  V. Undertaking Cervical cancer screening activity in Gobei Health Center(Bondo) and Jaramogi Oginga Odinga University of Science and Technology(Kisumu)
  VI. Undertaking Prevalence study on HPV and genotype distribution among HIV positive and HIV negative women in Lake Victoria Basin
  VII. Capacity building through training on HPV-DNA testing in Antwerp Belgium

- Project assistant in Objective 2: Community KAP, Healthcare worker preparedness and Hospital readiness in NCD management
  I. Development and Validation of data collection tools
  II. Monitoring and undertaking field work in Kisumu and Siaya County
  III. Part of Data analysis team
  IV. Development of monitoring and evaluation framework

PART-TIME LECTURER | JARAMOGI OGINGA ODINGA UNIVERSITY OF SCIENCE AND TECHNOLOGY- JAN 2019 TO-DATE

- Part time Lecturer at Jaramogi Oginga University of Science and Technology
- Subjects tutoring: Basic Microbiology, Principles of Epidemiology and Disease Control,

Healthcare financing, Drug and Substance abuse
- Molecular Biology training

MENTOR| AMREC COMMUNITY |JAN 2019 TO-DATE

Responsibilities

- Skill acquired in qualitative and quantitative Data Analysis.
- Working experience using SPSS, STATA, NVIVO, R
- Ability to develop and manipulate mobile apps like CommCare, ODK, and survey monkey.
- Perform visualization using Power BI
- Develop research Proposals, manuscript writing, systematic reviews, Meta-analysis

LAB SCIENTIST | AHERO SUB-COUNTY HOSPITAL | OCT 2014- DEC 2016

Responsibilities

- Clinical Laboratory Practice in Phlebotomy, virology, Hematology, bacteriology, pathology
- Participation in community health education and promotion
- Participation in community outreaches and diagnosis

- Other defined activities as agreed with the line supervisor

LEADERSHIP POSITIONS | KSCF | OCT 2014- DEC 2016

County Secretary

AWARDS/HONOR | KENYA GOVERNMENT | OCT 2014- DEC 2016

National Research Fund grant to carry out study; Risk factors and management of outcome of Premarital sex among adolescent youths attending secondary school in Kadibo Division, nyando Sub-County, Kisumu County, Kenya

GRANT AWARDED

WASH-HPV: Water, Hygiene and Sanitation and linkage to HPV infection.

Role: Local Principal Investigator

Amount: 70,000 Euros

Funder: VLIR-UOS

INTERNATIONAL CONFERENCE PROCEEDINGS

IPVS 2024 Conference in Edinburg UK November 2024.

1. Poster Presentation on Prevalence, genotype distribution of Potential HR/HR HPV among women attending selected facilities in Lake Victoria Basin
2. Poster Presentation on Prevalence and distribution of HPV-16 genetic variants among women in Africa: Systematic Review and Meta-analysis

Oncology Society Kenya, Eldoret August 2024

1. Oral Presentation on HPV infection pattern and viral load distribution: Implication for cervical cancer management

PUBLICATIONS

1. **Akinyi, Ivy**, Awandu, Shehu Shagari, Broeck, Davy Vanden, Pereira, Ana Rita, Redzic, Nina, & Bogers, Johannes. (2024). Prevalence and genotype distribution of potential high-risk and high-risk human papillomavirus among women attending selected reproductive health clinics in lake victoria basin-kenya: a cross-sectional study. *BMC Women's Health*, *24*(1), 468. doi:10.1186/s12905-024-03303-9
2. **Akinyi, I**., Ferreira, M.T., Lopez, R.V.M. *et al.* HPV-16 lineages among women in Africa: a systematic review with meta-analysis. *Syst Rev* **14**, 140 (2025). https://doi.org/10.1186/s13643-025-02891-3
3. **Akinyi I**, Ouma OJ, Ogutu S, Ogola E, Owenga J, Ayodo G, Omondi D, Awandu SS, Vanden Broeck D, Redzic N, Pereira AR, Bogers J. HPV infection patterns and viral load distribution: implication on cervical cancer prevention in Western Kenya. Eur J Cancer Prev. 2025 Jul 1;34(4):329-336. doi: 10.1097/CEJ.0000000000000920. Epub 2024 Aug 29. PMID: 39230048; PMCID: PMC12140554.
4. Ogony J, Mangeni J, Ayodo G, Menya D, **Akinyi I**, Oyugi B, et al. Correlation and epidemiologic distribution of emerging coinfections of Plasmodium falciparum and dengue virus among febrile children in malaria-endemic zones in western Kenya. IJID Regions. 2025;17:100737.
5. **Akinyi, I** & Ayodo, G. Risk factors and management of outcome among adolescent youths attending

secondary school in Kadibo Division, Nyando Sub county, *IJIR*

REFEREES

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Dr. Rita Ana Pereira, Algemeen Medisch Laboratorium (AML) - Sonic Healthcare Benelux 9, BE-2020, Antwerpen Belgium, TEL: +3233030800, rita.pereira@aml-lab.be

Dr. Dickens Omondi, JOOUST, School of Health Sciences, P.O.BOX 210-40601, BONDO, TEL: +254 722 385 291, omondisda@gmail.com,